THE MARKED UNIVERSAL PROPERTY OF BR, AND $\pi_4(S^3) = \mathbb{Z}/2$

Purpose and status. Step §2e of `doc/pi3-s2-postnikov-handoff-10.txt` (step 3 of handoff 9). We specify a class $s \in [S^3, BR]_*$ and prove that

$$s^* : [BR, Y]_* \to [S^3, Y]_*$$

is a bijection for every pointed 2-connected 4-type Y (Theorem 32); the two instances $Y = BR$ and $Y = \|S^3\|_4$ then give $BR \simeq \|S^3\|_4$ on classes and $\pi_4(S^3) \cong \pi_3(R) \cong \mathbb{Z}/2$ as groups (Theorem 33). *Draft, written before any code. Read by the user (28 Oct), corrections incorporated: Lemma 10 made explicit along* ***`ENat.e-nat`*** *with Lemma 9; Corollary 15 (Lemma 13 settles the numerical converse); the diagonal evaluation in Proposition 12.* Refereed *independently (a fresh reader given the note and the library, not the conclusion; 28 Oct): sound with repairs, none fatal; incorporated: a generic two-sided* ***`Conj`*** *(the single-degree* ***`Climb.conj-unconj`*** *does not apply), the lock bridges, the index of* ***`ConnPi1`*** *in Proposition 4, the direction of* ***`loop-homotopy`****, the obligations for π_4 on classes,* ***`Triple3`****, the input of the last step of* ***`Rec4`****,* ***`unique-loop1`****.* Numbered statements are meant to be declarations (§12 proposes names).

Assumption. The S^3 *axiom* (Axiom 1): the universal property of S^3, stated as a segment parameter telescope as `SphereAxiom` does for S^2, *not* as a Theory parameter. It is the only assumption beyond the foundations of the development (univalence, images; function extensionality derived). The S^2 axiom is *not* used: the generator of $\pi_3 K(\mathbb{Z}, 3)$ is built from the circle property (Definition 6), and everything about R, BR, κ is axiom-free (`doc/cross-term.pdf`, `H4Proof.h4`, `FibreH3`).

The decisive point. Constructing BR (`DeloopR`) proves that a delooping of $R = \mathrm{Fib}\bar{\kappa}$ *exists*; it says nothing about maps *out of* BR. A map $BR \to Y$ into a target presented as $Y \simeq \mathrm{Fib}K$, $K : K(A,3) \to_* K(B,5)$, exists over a given map to $K(A,3)$ exactly when an obstruction vanishes, and the obstruction is a class $[H] = \Phi_B(h) \in [K_2, K(B,4)]_* \cong B$. The fibre sequence of BR kills it only if h factors through $\rho : \mathbb{Z} \to \mathbb{Z}/2$, i.e. if $2h = 0$. So the proof must show: *for every target, the obstruction is 2-torsion.* The cross-term proof does not give this: it shows that *our* $\bar{\kappa}$ has a vanishing cross term because $2 = 0$ in $\mathbb{Z}/2$; the general statement "if $e_2; F; e_4^{-1}$ deloops, then $2g = 0$" is the converse of `doc/cross-term.pdf`, Proposition 21 (the handoff calls it 20), which is open there and is *not* used as established. We prove it here, as Lemma 13 ("necessity") and Corollary 15, by a route that avoids the obstacle of §7 of that note (normalising a cell); since $\mathsf{Unloop} f = \sum_{\hat{G}} (f = \Omega \hat{G})$, an inhabitant already supplies the map and the equality Lemma 13 needs:

- the obstruction H is the looping of a map $K(\mathbb{Z},3) \to_* K(B,5)$ *by construction* (it is $\mathrm{unconj}(\Omega M)$ for $M = G; K$), so no cell of `Unloop` has to be normalised;
- ΩM commutes with squaring of loops (`loop-mult`), and squaring is $K(2,2)$ on the source and $K(2_B, 4)$ on the target (`Doubling`), so $K(2,2); H$ and $H; K(2_B, 4)$ are pointwise equal; this is a statement about underlying maps into $K(B,4)$, where pointings are free (merely);
- a computation with the bipointed classification (Theorem B of `doc/h4.pdf`) gives $[K(2,2); \kappa] = [\kappa; K(4,4)]$ (Proposition 12: $(2x)^2 = 4x^2$), hence $\Phi_B(4h) = \Phi_B(2h)$ and $2h = 0$ by injectivity of Φ_B.

The only new ingredient of the computation is the naturality of the invariant $\mathcal{E}$ under precomposition in the *second* variable (Lemma 10); first-variable doubling is `ETwo.e-two`.

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Why at the level of BR, not of R. One could loop everything ($[BR, Y]_* \cong [R, \Omega Y]_*$, Lemma 5) and run `MapFib` with source R. But then the target is only a 1-connected 3-type ΩY *with the property* of being a loop space, and the obstruction would have to be extracted from an equivalence $\Omega Y \simeq \mathrm{Fib}k$, which is the open converse in disguise. Presenting the target one level up (Theorem 31) makes its k-invariant a map $K : K(A,3) \to_* K(B,5)$, so every obstruction is a looping by construction. The proof is therefore `MapFib` with every index raised by one, plus Lemma 13 in the proof of (E) for the source BR (Proposition 22).

Restricted coefficients. The coefficient map of BR is $\Psi_B : \mathrm{Hom}(\mathbb{Z}/2, B) \to [K_2, K(B,4)]_*$, $g \mapsto [\bar\kappa; K(g,4)] = \Phi_B(\rho; g)$. It is injective (Proposition 17), which is what `FibreH3` needs (so $[R, K(B,3)]_* = 0$, Proposition 18); and its image contains every looping class (Lemmas 13, 16), which is what (E) needs. The unrestricted interface (`H4`, `FibreH3.FibAt`, `MapFib.SourceP`) is not instantiated with $\bar\kappa$; the restricted statements are written out below.

1. Conventions

As in `doc/mapping-fib.pdf`: `isTrunc n` is "$(n-2)$-truncated", `isConn n` is "$(n-2)$-connected"; $K^A_m = K(A,m)$ (`EMMaps.KK`), $K_m = K^{\mathbb{Z}}_m$, $K'_m = K^{\mathbb{Z}/2}_m$; $e_m : \Omega K^A_{m+1} \simeq_* K^A_m$; $F;G$ is `ptcomp F G`; 0 is `ptconst`; $[F]$ is `hcl F`, $a;b$ on classes is `compC`; equality of classes is mere equality of pointed maps (`effective`, `relates`). $\mathrm{conj}_m F = e_m; F; e_m^{-1}$ and $\mathrm{unconj}_m(f) = e_m^{-1}; f; e_m$ (`FibreH3.Climb.unconj`; conj ∘ unconj = id, `conj-unconj`). In §5 and Proposition 22, conj is used with *different degrees and groups* on the two sides ($\mathrm{conj}F = e_2; F; e_4^{-1} : \Omega K_3 \to_* \Omega K_5^B$ for $F : K_2 \to_* K_4^B$), which the single-degree `Lev.conj`, `Climb.conj-unconj`, `H2.unconj-conj` do not cover. A generic `Conj[eA,eB]` for two arbitrary pointed equivalences $e_A : \Omega X \simeq_* A$, $e_B : \Omega Y \simeq_* B$ is written first, with `conj-unconj`, `unconj-conj` (proofs as `Climb.conj-unconj`) and composition $\mathrm{conj}_{A,C}(F;G) = \mathrm{conj}_{A,B}F; \mathrm{conj}_{B,C}G$ (`KFun.conj-comp`, already generic); every conj below is an instance; $\Omega K(g, m{+}1) = \mathrm{conj}_m K(g,m)$ (`loopK`). $\Phi_B(h) = [\kappa; K(h,4)]$ for $h : \mathrm{Hom}(\mathbb{Z}, B)$ (`KZ2Cohomology.Phi`), a bijection (`H4Proof.h4`). For homomorphisms, $g; h$ is `hcomp g h` $= h \circ g$; $K(g;h,m) = K(g,m); K(h,m)$ (`KFun.Comp.k-comp`). 2_A, 0, ρ as in `Z2`; $4_{\mathbb{Z}} = 2_{\mathbb{Z}}; 2_{\mathbb{Z}}$.

Levels: K^A_m is $\mathsf{isConn}(m{+}1)$ and $\mathsf{isTrunc}(m{+}2)$. A *pointed 2-connected 4-type* is a pointed Y with $\mathsf{isConn}\,4$ and $\mathsf{isTrunc}\,6$. From `doc/cross-term.pdf`: $\bar\kappa = \kappa; K(\rho,4)$, $R = \mathrm{Fib}\bar\kappa$ (`isConn 3`, `isTrunc 5`), $\hat F : K_3 \to_* K'_5$ with $\gamma : e_2; \bar\kappa; e_4^{-1} = \Omega\hat F$, $BR = \mathrm{Fib}\hat F$ pointed by $b_0 = (*, \hat F_0)$, $p = \mathrm{fpr} : BR \to_* K_3$, $\phi : \Omega BR \simeq_* R$ (`omega-BR`); BR is $\mathsf{isConn}\,4$, $\mathsf{isTrunc}\,6$.

2. The S^3 axiom and the 4-truncation of S^3

Axiom 1 (S^3)**.** A type S^3 with base and $\mathsf{surf} : \Omega^3(S^3, \mathsf{base})$ such that, for every X of the universe of S^3, $\mathrm{ev}_3 : (S^3 \to X) \to (x : X) \times \Omega^3(X,x)$, $f \mapsto (f\,\mathsf{base}, \mathsf{ap}^3_f \mathsf{surf})$ is an equivalence. A segment parameter telescope (`S3, base, surf, ump3`), as `SphereAxiom.Sphere`; not a Theory parameter.

Lemma 2 (as `SphereAxiom`)**.** $\mathrm{ev} : \mathrm{Map}_*(S^3, X) \to \Omega^3 X$, $F \mapsto (\Omega^3 F)(\mathsf{surf})$*, is an equivalence (`pointed-up`:* Σ *over `ump3` and contraction of the singleton, with* $\mathrm{ev}(f,\pi) = \pi_* \mathsf{ap}^3_f \mathsf{surf}$ *by J), and* $\mathrm{ev}(F;G) = (\Omega^3 G)(\mathrm{ev}F)$ *(`loop-comp` three times).*

Lemma 3 (S^3 is 2-connected)**.** S^3 *is* $\mathsf{isConn}\,4$.

Proof. As `SphereTrunc.Simply`, one level up: for Y `isTrunc 4`, $\Omega^3(Y,y)$ is contractible, so evaluation at base is an equivalence $(S^3 \to Y) \simeq Y$; hence $V = \|S^3\|_2$ (`TO three`) is contractible (the identity and the constant map agree after the unit), and S^3 is the fibre of the unit over $|\mathsf{base}|$, which is $\mathsf{isConn}\,4$. □

Proposition 4 (the 4-truncation)**.** $T = \|S^3\|_4$ *(`TO five`), pointed by* $t_0 = |\mathsf{base}|$*, with* $\eta = (|-|, \mathsf{refl}) : S^3 \to_* T$*, satisfies: T is* $\mathsf{isTrunc}\,6$ *and* $\mathsf{isConn}\,4$*; $\eta^* : \mathrm{Map}_*(T,Z) \to \mathrm{Map}_*(S^3,Z)$ is an*

equivalence for Z `isTrunc 6` (`PtUP.up`); and $\pi_4\eta : \pi_4 S^3 \to \pi_4 T$ is a group isomorphism. Here $\pi_4(X, x_0) = \pi_3(\Omega X, \mathsf{refl})$ by definition, and $\pi_4 F = \pi_3(\Omega F)$.

Proof. As `SphereTrunc.Theorem15`: connectedness by `conn-up` from the fibres of the unit and Lemma 3; the fibres of $\Omega^k\eta$ are $\mathsf{isConn}(6-k)$ (`loop-conn`); `ConnPi1` for $\Omega^3\eta$ takes the connectivity of the fibres of its looping $\Omega^4\eta$, which are $\mathsf{isConn}\,2$ (`loop-conn` four times), and makes $\pi_1(\Omega^3\eta) = \pi_4\eta$ bijective. □

3. $K(\mathbb{Z}, 3)$ is the 3-truncation of S^3

Lemma 5 (looping is an equivalence)**.** *For X `isConn (c+2)` and Y `isTrunc (2c+2)`, $\Omega : \mathrm{Map}_*(X, Y) \to \mathrm{Map}_*(\Omega X, \Omega Y)$ is an equivalence (`Over.cor11`); used with $c = 2$ ($X = K_3$ or BR, Y a 3-type or 4-type) and $c = 1$ ($X = K_2$, `MapFib.SourceS.LoopEq`). For X `isConn 4` and Y `isTrunc 7`, Ω is an embedding: its fibres $\mathsf{Unloop}f$ are propositions (`Levels[2,1].cor10`).*

Definition 6 (σ_3)**.** Let $g : \Omega K_1$ be the circle generator, $c_3 = (\Omega^2 e_2^{-1})((\Omega e_1^{-1})(g)) : \Omega^3 K_3$, and $\sigma_3 = \mathrm{ev}^{-1}(c_3) : S^3 \to_* K_3$.

Proposition 7. *For every 3-type (Z, z_0) (`isTrunc 5`), $\sigma_3^* : \mathrm{Map}_*(K_3, Z) \to \mathrm{Map}_*(S^3, Z)$ is an equivalence; hence $[\sigma_3]^* : [K_3, Z]_* \to [S^3, Z]_*$ is a bijection.*

Proof. Let $\Psi_3(G) = \mathrm{ev}^\circ\big(e_1^{-1}; \Omega(e_2^{-1}; \Omega G)\big)$, where $\mathrm{ev}^\circ : \mathrm{Map}_*(K_1, \Omega^2 Z) \simeq \Omega^3 Z$, $N \mapsto (\Omega N)(g)$, is the circle property. Ψ_3 is a composite of five equivalences: Ω (Lemma 5, $c = 2$; Z is $\mathsf{isTrunc}\,6$ by `trunc-suc`), $e_2^{-1}; -$ (`pre-pteq`), Ω ($c = 1$; ΩZ is $\mathsf{isTrunc}\,4$), $e_1^{-1}; -$, ev°. By Lemma 2 and `loop-comp`,

$$\mathrm{ev}(\sigma_3; G) = (\Omega^3 G)(c_3) = \big(\Omega^2(e_2^{-1}; \Omega G)\big)((\Omega e_1^{-1})g) = \big(\Omega(e_1^{-1}; \Omega(e_2^{-1}; \Omega G))\big)(g) = \Psi_3(G).$$

So $\mathrm{ev} \circ \sigma_3^* = \Psi_3$ and σ_3^* is an equivalence (`cancel-equiv`); classes by `pre-equiv`. The S^2 axiom is not used. □

4. Level-3 triples and the lifting lemma

A *level-3 triple* is (A, B, K) with $K : K_3^A \to_* K_5^B$. Its fibre $X = \mathrm{Fib}K$, pointed by $(*, K_0)$, with $q = \mathrm{fpr} : X \to_* K_3^A$, is $\mathsf{isConn}\,4$ (fibres $K\,y = *$ are $\mathsf{isConn}\,5$; `FibConn`) and $\mathsf{isTrunc}\,6$. ΩK_5^B is $\mathsf{isConn}\,5$, $\Omega^3 K_5^B$ is $\mathsf{isConn}\,3$, $\Omega^4 K_5^B$ is $\mathsf{isConn}\,2$, ΩX is $\mathsf{isConn}\,3$; all connected.

Theorem 8 (lifting, `MapFib` Theorem 6 one level up)**.** *Let (Q, q_0) be pointed and (A, B, K) a level-3 triple. Assume (E) for every $F : Q \to_* K_3^A$, merely $F; K = 0$; and (U) every $d : Q \to_* \Omega K_5^B$ is merely 0. Then $q_* : [Q, X]_* \to [Q, K_3^A]_*$ is a bijection.*

Proof. The text of `doc/mapping-fib.pdf`, §2, with K_2^A, K_4^B replaced by K_3^A, K_5^B: existence from `Fibres.Lift`; uniqueness from (U) through (U_0) (ΩK_5^B connected), `cancel-snd`, and pointing irrelevance (ΩX connected). Only connectedness of ΩK_5^B and ΩX is used. Formally, `MapFib.Lifting` is stated at level 2, so it is either generalised to any $k : (Y_0, y_0) \to_* (W, w_0)$ with these two connectedness hypotheses, or copied (§12). □

5. Doubling detects the coefficient

This section is the necessity argument. Write $M_Y = \mathrm{Map}_*(K_2, Y)$, $c_2 : K_2 \to_* M_{K_4}$ the cup map, $\kappa = \Delta c_2$, $\iota = \mathcal{E}(c_2) : K_2 \to_* \Omega^2 K_4$ (`cup-norm`). For $G : K_2 \to_* K_2$, let $G^\sharp : M_Y \to_* M_Y$, $m \mapsto G; m$, pointed by `comp-const-r`.

Lemma 9 (the pointing of $G^\sharp$, evaluated)**.** *For $G : K_2 \to_* K_2$ and $x : K_2$, $\mathsf{ap}_{\mathrm{ev}_x}(G_0^\sharp) = \mathsf{refl}$, where $\mathrm{ev}_x(m) = m_1(x)$ and $G_0^\sharp =$ `comp-const-r`$\,G : G; 0 = 0$.*

Proof. By definition, `comp-const-r` is $\mathsf{ap}_{\lambda e.(\lambda_.y_0,\, e)}$ of `ap-const` : $\mathsf{ap}_{\lambda_.y_0} G_0 = \mathsf{refl}$: it varies only the pointing and keeps the underlying constant map fixed. So $\mathsf{ap}_{\mathrm{ev}_x}(G_0^\sharp) = \mathsf{ap}_{\lambda e.\, y_0}(\texttt{ap-const}\ldots)$ (`ap-ap`), which is refl by `ap-const`. □

Lemma 10 ($\mathcal{E}$ in the second variable)**.** *For $c : K_2 \to_* M_Y$, $G : K_2 \to_* K_2$ and $x : K_2$, $\mathcal{E}(c; G^\sharp)(x) = \mathcal{E}(c)(G_1 x)$ in $\Omega^2 Y$.*

Proof. The pattern of `ENat.e-nat` (`H4Bilinear`), with precomposition in place of postcomposition; only one target Y, so one circle instance and no `g-eq`. Write $\phi(c) = e_1^{-1}; \Omega c; \mathsf{ev}^{L4} : K_1 \to_* N$, $N = \mathrm{Map}_*(K_2, \Omega Y)$, and $\mathrm{Ex}_x = (F \mapsto F_1 x, \mathsf{refl}) : N \to_* \Omega Y$. *(1)* $\mathcal{E}(c)(x) = (\Omega(\phi(c); \mathrm{Ex}_x))(g)$ (`E-loop`: `ev-ap`, `evPt-loop`, `loop-comp`); likewise $\mathcal{E}(c; G^\sharp)(x) = (\Omega A_1)(g)$ with $A_1 = \phi(c; G^\sharp); \mathrm{Ex}_x$, and $\mathcal{E}(c)(G_1 x) = (\Omega A_2)(g)$ with $A_2 = \phi(c); \mathrm{Ex}_{G_1 x}$; both $A_i : K_1 \to_* \Omega Y$. *(2) An unpointed homotopy* $A_1(k) = A_2(k)$ for $k : K_1$. With $P = (\Omega c)(e_1^{-1} k)$, a loop of M_Y at 0: $A_1(k) = \mathsf{ap}_{\mathrm{ev}_x}((\Omega G^\sharp) P)$ (`ev-ap`, `loop-comp`) $= \mathsf{ap}_{\mathrm{ev}_x}\big((G_0^\sharp)^{-1} \cdot \mathsf{ap}_{G^\sharp} P \cdot G_0^\sharp\big)$ (`loop-ap`) $= \mathrm{conj}_{\mathsf{ap}_{\mathrm{ev}_x} G_0^\sharp}\big(\mathsf{ap}_{\mathrm{ev}_x \circ G^\sharp} P\big)$ (`ap-conj`, `ap-ap`) $= \mathsf{ap}_{\mathrm{ev}_{G_1 x}} P$ (Lemma 9, `conj-refl-loop`, and $\mathrm{ev}_x \circ G^\sharp \equiv \mathrm{ev}_{G_1 x}$ by definition of `ptcomp`) $= A_2(k)$ (`ev-ap`). This is the analogue of `nat-pt`/`hh`; no pointing is compared. *(3) Looping the homotopy.* `LoopConj.LoopHomotopy` for A_1, A_2 and this homotopy gives a 2-loop $\mathsf{c} : \Omega^2 Y$ with $(\Omega A_2)(g) = \mathrm{conj}_{\mathsf{c}}((\Omega A_1)(g))$ in $\Omega(\Omega Y)$ (`loop-homotopy`); the conjugation is removed by Eckmann–Hilton (`EH[Y,y0].eh`, `trans-cancel`), exactly as in `e-nat`. Hence $\mathcal{E}(c; G^\sharp)(x) = \mathcal{E}(c)(G_1 x)$. □

Lemma 11. *$\iota(K(2,2)\, y) = \iota y \cdot \iota y$ in $\Omega^2 K_4$, for all $y : K_2$.*

Proof. ι is the inverse of $j = \Omega e_3; e_2 : \Omega^2 K_4 \simeq_* K_2$ (`iota-eq`); apply the injective j. $j(\iota y \cdot \iota y) = e_2((\Omega e_3)(\iota y) \cdot (\Omega e_3)(\iota y))$ (`loop-mult`) $= K(2,2)(e_2((\Omega e_3)(\iota y)))$ (`Dbl.dbl` at $m = 2$, $A = \mathbb{Z}$: $e_2(p \cdot p) = K(2,2)(e_2 p)$, by `sec-eq`) $= K(2,2)(j(\iota y)) = K(2,2) y = j(\iota(K(2,2) y))$. □

Proposition 12 ($(2x)^2 = 4x^2$)**.** *$K(2,2); c_2; K(2,2)^\sharp = c_2; K(4,4)_*$ in $\mathrm{Map}_*(K_2, M_{K_4})$; hence $K(2,2); \kappa$ and $\kappa; K(4,4)$ are pointwise equal, and $[K(2,2)]; [\kappa] = [\kappa]; [K(4,4)]$.*

Proof. (`ETwo.e-two` and `Dbl.dbl` are stated with the unlocked `KMap[..,twoH].K` and `KT.conj`; the locked `Lev` versions used here are bridged by `K-def`/`conj-def`, and `ptcomp-assoc` puts $K(2,2); c_2; K(2,2)^\sharp$ in the shape $K(2,2); c'$ expected by `e-two`; the same bridges are used in Lemmas 11 and 13.) By Theorem B at $B = \mathbb{Z}$ (`BilAt[Zab].theta-inj`) and `pi-set` (`sK2`, `cK2`, as in `Kill.kill`) it suffices that the two $\mathcal{E}$ are pointwise equal. With $w = \iota y$: $\mathcal{E}(K(2,2); c_2; K(2,2)^\sharp)(y) = v \cdot v$ with $v = \mathcal{E}(c_2; K(2,2)^\sharp)(y)$ (`ETwo.e-two`), and $v = \iota(K(2,2) y) = w \cdot w$ (Lemmas 10, 11). On the other side, $\mathcal{E}(c_2; K(4,4)_*)(y) = (\Omega^2 K(4,4))(w)$ (`ENat.e-nat`), $K(4,4) = K(2,4); K(2,4)$ (`k-comp`), and $(\Omega^2 K(2,4))(u) = u \cdot u$ (`Dbl.dbl` at $m = 3$ gives $\Omega K(2,4) = \mathsf{sq}$, then `loop-sq`), so it is $(w \cdot w) \cdot (w \cdot w)$. For the diagonal: by definition $\Delta(K(2,2); c; K(2,2)^\sharp)(x) = K(2,2)^\sharp\big(c(K(2,2)x)\big)(x) = c(K(2,2)x)(K(2,2)x)$: the precomposition already moves the second argument. For $c = c_2$ this is $\kappa(K(2,2)x)$, and $\Delta(c_2; K(4,4)_*)(x) = K(4,4)(\kappa x)$; evaluate the equality of bipointed maps at x, then at x. Pointings into K_4 are merely irrelevant (`pi-mere`; ΩK_4 is connected), which gives the classes. □

Lemma 13 (necessity)**.** *Let $\hat{G} : K_3 \to_* K_5^B$, $H = \mathrm{unconj}(\Omega \hat{G}) : K_2 \to_* K_4^B$, and $h : \mathrm{Hom}(\mathbb{Z}, B)$ with $\Phi_B(h) = [H]$. Then $h(1) + h(1) = 0$.*

Proof. (i) Pointwise. $\mathrm{conj} H = \Omega \hat{G}$ (`conj-unconj`). By `Dbl.dbl`, $\mathsf{sq} = \mathrm{conj}_2 K(2,2)$ on ΩK_3 and $\mathsf{sq} = \mathrm{conj}_4 K(2_B, 4)$ on ΩK_5^B. So $\mathrm{conj}(K(2,2); H) = \mathrm{conj}_2 K(2,2); \mathrm{conj} H = \mathsf{sq}; \Omega \hat{G}$ and $\mathrm{conj}(H; K(2_B,4)) = \Omega \hat{G}; \mathsf{sq}$ (associativity, $e^{-1}; e = \mathrm{id}$ in the middle). These have pointwise equal underlying maps, since $(\Omega \hat{G})(p \cdot p) = (\Omega \hat{G}) p \cdot (\Omega \hat{G}) p$ (`loop-mult`). Applying e_4 and e_2^{-1} (`sec-eq`), $K(2,2); H$ and $H; K(2_B, 4)$ have pointwise equal underlying maps; by `pi-mere` (ΩK_4^B connected) they are merely equal, so $[K(2,2)]; [H] = [H]; [K(2_B,4)]$. *(ii) Classes.* Substitute $[H] = [\kappa]; [K(h,4)]$ (`Lev.K-def` as in `H4Proof.phi-diag`). Left: $[K(2,2)]; [\kappa]; [K(h,4)] =$

$[\kappa];[K(4,4)];[K(h,4)] = \Phi_B(4_{\mathbb{Z}};h)$ (Proposition 12, `comp-assoc`, `k-comp`). Right: $[\kappa];[K(h,4)];[K(2_B,4)] = \Phi_B(h;2_B)$. By injectivity of Φ_B, $h \circ 4 = 2_B \circ h$; at 1, with $4_{\mathbb{Z}}(1) = (1+1)+(1+1)$ (`twoH`, `hcomp`) and h a homomorphism, $(h1 + h1) + (h1 + h1) = h1 + h1$; cancelling in B, $h1 + h1 = 0$. □

Remark 14. This is exactly where a general target differs from BR: for $B = \mathbb{Z}/2$ the conclusion is Lemma `two-zero`, and Proposition 12 is not needed. For $Y = \|S^3\|_4$ the group $B = \pi_4$ is the unknown, so Lemma 13 cannot be avoided. No cell of Unloop is normalised, and no bipointed map is shown to vanish from a pointwise vanishing; only classes into K_4^B are compared.

Corollary 15 (the numerical converse)**.** *For $h : \mathrm{Hom}(\mathbb{Z}, B)$ let $f_h = \mathrm{conj}(\kappa; K(h,4)) : \Omega K_3 \to_* \Omega K_5^B$. If $\mathsf{Unloop} f_h$ is inhabited, then $2h = 0$. So $e_2;\kappa;K(h,4);e_4^{-1}$ deloops if and only if $2h = 0$: for "if", $h = \rho; g$ (Lemma 16), so $\kappa;K(h,4) = \bar\kappa;K(g,4)$ (`k-comp`) and $f_h = \mathrm{conj}(\bar\kappa;K(g,4)) = \Omega(\hat F;K(g,5))$, as in the proof of Proposition 22.*

Proof. By definition $\mathsf{Unloop} f = \sum_{\hat G}(f = \Omega\hat G)$, so an inhabitant is a pair $(\hat G,\ \pi : f_h = \Omega\hat G)$: exactly the data of Lemma 13, without any normalisation of a cell. $H = \mathrm{unconj}(\Omega\hat G) = \mathrm{unconj}(f_h) = \kappa;K(h,4)$ (along π^{-1}, and $\mathrm{unconj} \circ \mathrm{conj} = \mathrm{id}$, `pt-inv-l/r`, as `conj-unconj`), so $[H] = \Phi_B(h)$, and Lemma 13 gives $h1 + h1 = 0$, i.e. $2_B \circ h = 0$ (homomorphisms out of $\mathbb{Z}$ are determined at 1). □

What remains open from §7 of `doc/cross-term.pdf` is only a *route*: normalising the cell given by `lemma9` at (w_0, f_1), and the general cross-term formula $\tilde\delta_F = c_2; K(2g,4)_*$. Neither is used, and neither is needed for the numerical statement.

Lemma 16 (factorisation through ρ)**.** *If $h : \mathrm{Hom}(\mathbb{Z}, B)$ and $h1 + h1 = 0$, then $h = \rho; g$ for $g : \mathrm{Hom}(\mathbb{Z}/2, B)$, $g(\mathsf{false}) = 0$, $g(\mathsf{true}) = h1$.*

Proof. g is a homomorphism by cases (the case $\mathsf{true} + \mathsf{true}$ uses the hypothesis). $g(\rho x) = h(x)$ by induction on $x : \mathbb{Z}$ (`suc`, `pred` as `rho-hom`); homomorphisms are equal when their maps are (`hom-path`). □

Proposition 17 (the restricted coefficient map)**.** *$\Psi_B : \mathrm{Hom}(\mathbb{Z}/2, B) \to [K_2, K_4^B]_*$, $\Psi_B(g) = [\bar\kappa; K(g,4)]$, satisfies $\Psi_B(g) = \Phi_B(\rho; g)$ and is injective. Every class $[\mathrm{unconj}(\Omega\hat G)]$ lies in its image.*

Proof. $\bar\kappa;K(g,4) = \kappa;K(\rho,4);K(g,4) = \kappa;K(\rho;g,4)$ (`k-comp`; the same locked `Lev` $K(\rho,4)$ as in $\bar\kappa$). Injective: $\rho;g = \rho;g'$ gives $g(\mathsf{true}) = g'(\mathsf{true})$ at $x = 1$, and $g(\mathsf{false}) = 0 = g'(\mathsf{false})$. Image: Lemmas 13 and 16. (Conversely every $\bar\kappa;K(g,4)$ is a looping class, of $\hat F;K(g,5)$; not needed.) □

6. The source BR

Proposition 18 ($[R, K_3^B]_* = 0$)**.** *Every $d : R \to_* \Omega K_4^B$ is merely 0.*

Proof. `doc/fibre-h3.pdf`, Theorem 10, for the triple $(\mathbb{Z}, \mathbb{Z}/2, \bar\kappa)$ in place of $(\mathbb{Z}, \mathbb{Z}, \kappa)$: the fibre over the base point is $\Omega K_4'$; the restriction $\mathrm{res}(d)$ deloops to $G : K_4' \to_* K_4^B$ (Corollary 11, $c = 2$); the transgression gives $[\bar\kappa; G] = [0]$; `Climb[Z2ab,B].surj4` gives $G = K(g,4)$ and $0 = K(g_0,4)$ with $g, g_0 : \mathrm{Hom}(\mathbb{Z}/2, B)$; so $\Psi_B(g) = \Psi_B(g_0)$, $g = g_0$ (Proposition 17), $G = 0$, $\mathrm{res}(d) = 0$, and exactness ((H3) for the base K_2) gives d merely 0. The proof of `FibreH3` uses of κ only the injectivity of $g \mapsto [\kappa; K(g,4)]$; it is restated with a fibre coefficient group C, a map $k : K_2 \to_* K_4^C$ and that injectivity as hypothesis, and instantiated at $(C, k) = (\mathbb{Z}/2, \bar\kappa)$ with Proposition 17. □

Proposition 19 ((U) for BR)**.** *Every $d : BR \to_* \Omega K_5^B$ is merely 0.*

Proof. $\Omega : \mathrm{Map}_*(BR, \Omega K_5^B) \to \mathrm{Map}_*(\Omega BR, \Omega^2 K_5^B)$ is an equivalence (Lemma 5, $c = 2$: ΩK_5^B is $\mathsf{isTrunc}\, 6$). $\phi^{-1}; \Omega d : R \to_* \Omega^2 K_5^B$ is merely 0 by Proposition 18, transported along $\Omega e_4 : \Omega^2 K_5^B \simeq_* \Omega K_4^B$ (`ptEqInd`); so merely $\Omega d = 0 = \Omega 0$ (`pre-pteq`, `loop-const`), and $d = 0$ by injectivity of Ω. □

Lemma 20 (p^*, as `MapFib.SourceP.PStar`)**.** *The fibres of $p_1 : BR \to K_3$ are* $\mathsf{isConn}\,5$ *(Lemma 7 of `doc/mapping-fib.pdf`, `fib-fst`; K_5' is* $\mathsf{isConn}\,6$*). For every A, $p^* : [K_3, K_3^A]_* \to [BR, K_3^A]_*$ is a bijection, and every $F : BR \to_* K_3^A$ is merely $p; G_F$ with $G_F = (\mathrm{ext}\, F_1, \zeta_F)$ (`ConnectedMaps.Elim`, $m = 5$; K_3^A is* $\mathsf{isTrunc}\,5$*; ΩK_3^A connected).*

Lemma 21 (the fibre sequence)**.** *Merely $p; \hat{F} = 0$ ($\lambda w.w_2$ is a homotopy; $\Omega K_5'$ connected).*

Proposition 22 ((E) for BR)**.** *For every level-3 triple (A, B, K) and $F : BR \to_* K_3^A$, merely $F; K = 0$.*

Proof. The goal is a proposition. Let $M = G_F; K : K_3 \to_* K_5^B$, so merely $F; K = p; M$ (Lemma 20, `ptcomp-assoc`). Let $H = \mathrm{unconj}(\Omega M)$ and $h = \Phi_B^{-1}[H]$ (surjectivity of Φ_B). By Lemma 13, $h1 + h1 = 0$; by Lemma 16, $h = \rho; g$; so $[H] = \Phi_B(\rho; g) = [\bar{\kappa}; K(g,4)]$ (Proposition 17) and merely $H = \bar{\kappa}; K(g,4)$ (`effective`). Then, merely,

$$\Omega M = \mathrm{conj} H = \mathrm{conj}(\bar{\kappa}; K(g,4)) = (e_2; \bar{\kappa}; e_4^{-1}); (e_4; K(g,4); e_4^{-1}) = \Omega\hat{F}; \Omega K(g,5) = \Omega(\hat{F}; K(g,5))$$

(`conj-unconj`; $e_4^{-1}; e_4 = \mathrm{id}$ on K_4'; γ; `loopK`; `loop-comp`). By Lemma 5 (K_3 is $\mathsf{isConn}\,4$, K_5^B is $\mathsf{isTrunc}\,7$), $\mathsf{Unloop}(\Omega M)$ is a proposition containing (M, refl) and $(\hat{F}; K(g,5), \ldots)$; so merely $M = \hat{F}; K(g,5)$ (this is `LoopCalc.unique-loop1` at $c = 2$, which can be cited directly), and $p; M = (p; \hat{F}); K(g,5) = 0; K(g,5) = 0$ merely (Lemma 21, `comp-const-l`). □

Corollary 23. *For every level-3 triple, $q_* : [BR, \mathrm{Fib}K]_* \to [BR, K_3^A]_*$ is a bijection (Theorem 8, Propositions 22, 19).*

7. The source S^3 and the marked class

Lemma 24. *If $\Omega^3(Z, z_0)$ is connected, any two $F, F' : S^3 \to_* Z$ are merely equal (*ev *is injective, Lemma 2).*

Corollary 25. *(E) and (U) hold for $Q = S^3$ and every level-3 triple ($\Omega^3 K_5^B$ and $\Omega^4 K_5^B$ are connected). So $q_* : [S^3, \mathrm{Fib}K]_* \to [S^3, K_3^A]_*$ is a bijection; in particular $p_* : [S^3, BR]_* \to [S^3, K_3]_*$ (the triple $(\mathbb{Z}, \mathbb{Z}/2, \hat{F})$).*

Definition 26 (the marked class)**.** $s = p_*^{-1}[\sigma_3] \in [S^3, BR]_*$, so $s; [p] = [\sigma_3]$. A class, obtained without choice.

Remark 27. Under $[S^3, BR]_* \cong \pi_3 BR \cong \pi_2 R \cong \pi_2 K_2 \cong \mathbb{Z}$ the class s goes to 1; this is not used: what is used is Proposition 28, the property that characterises the generator.

Proposition 28. *For every A, $s^* : [BR, K_3^A]_* \to [S^3, K_3^A]_*$ is a bijection.*

Proof. $s^* \circ p^* = [\sigma_3]^*$ (`comp-assoc`); p^* and $[\sigma_3]^*$ are bijections (Lemma 20, Proposition 7, K_3^A a 3-type); `cancel-r`. □

8. The mapping theorem

Theorem 29. *For every level-3 triple (A, B, K), $s^* : [BR, \mathrm{Fib}K]_* \to [S^3, \mathrm{Fib}K]_*$ is a bijection.*

Proof. The square $q_* \circ s^* = s^* \circ q_*$ (`comp-assoc`) with top Corollary 23, right Proposition 28, bottom Corollary 25; `cancel-equiv`. □

Corollary 30 (transfer)**.** *If merely $\mathrm{Fib}K \simeq_* (Y, y_0)$ for some level-3 triple, then $s^* : [BR, Y]_* \to [S^3, Y]_*$ is a bijection (`ptEqInd`; the statement is a proposition, `isEquiv-prop`).*

9. Postnikov presentation of 2-connected 4-types

Theorem 31 (`Presentation` one level up)**.** *Every pointed 2-connected 4-type (Y, y_0) has merely a pointed equivalence* $\mathrm{Fib}K \simeq_* (Y, y_0)$ *for a level-3 triple (A, B, K).*

Proof. The text of `doc/postnikov-presentation.pdf`, §§3–6, with indices raised, except for the base, which is recognised instead of compared with a sphere:

(1) $V = \|Y\|_3$ (`TO four`, $\mathsf{isTrunc}\,5$), $u : Y \to_* V$, fibres E_z of u $\mathsf{isConn}\,5$ (`TruncOp.conn`) and $\mathsf{isTrunc}\,6$; V is $\mathsf{isConn}\,4$ (`conn-up`).
(2) *Base.* `Rec` (Proposition 7 there) gives $e_V : K_3^A \simeq_* V$, $A = \Omega^3 V$ (`O3`).
(3) *Fibre.* `Rec4`: for F $\mathsf{isConn}\,5$ and $\mathsf{isTrunc}\,6$, $K_4^B \simeq_* F$ with $B = \Omega^3(\Omega F)$: `Rec` for ΩF gives $K_3^B \simeq_* \Omega F$, and one more application of `deloop-equiv` with Corollary 11 at $c = 3$ (K_4^B is $\mathsf{isConn}\,5$; F is $\mathsf{isTrunc}\,6$, hence $\mathsf{isTrunc}\,8$), whose input is the pointed map $f_4 = e_3; G_3 : \Omega K_4^B \to_* \Omega F$ (G_3 the equivalence of `Rec`) together with the fact that $(f_4)_1$ is an equivalence (a composite). Applied to $F_0 = E_{v_0}$.
(4) *Homogeneity.* $Q\,t = \big(K_4^B \simeq_* (E_{ut}, (t, \mathsf{refl}))\big)$ is a set (`cor5-equiv`, $t = 1$, $c = 4$); a section through $Q\,t_0 \ni e_{F_0}$ exists (`lemma4`, $t = 0$; Y is $\mathsf{isConn}\,3$); moved along paths, $\mu_z : \mathrm{Mu}(E_z)$.
(5) *Classifying map.* K_5^B is the type of K_4^B-torsors by definition of `EM` (the stage $n = 2$ of $K(B, n{+}3) = T\,K(B, n{+}2)$); $k_1'(z) = \langle E_z, \mu_z \rangle$, $\mathrm{Fib}k' \simeq Y$ as Proposition 13 there; $K = e_V; k'$, $\mathrm{Fib}K \simeq Y$ (`sigma-base`); merely pointed (Y connected).

The new pieces are `Rec4` and the level-2 stage of the torsor identification (`K5-def`, `pt5-def`, τ); the rest is the same text. □

10. The universal property and $\pi_4(S^3)$

Theorem 32 (marked universal property of BR)**.** *For every pointed 2-connected 4-type (Y, y_0), $s^* : [BR, Y]_* \to [S^3, Y]_*$ is a bijection.*

Proof. Theorem 31 and Corollary 30. □

Only two instances are used. $Y = BR$: $BR = \mathrm{Fib}\hat{F}$ literally, so Theorem 29 for $(\mathbb{Z}, \mathbb{Z}/2, \hat{F})$ applies without Theorem 31; in Proposition 22 the coefficient group is $\mathbb{Z}/2$ and Lemma 13 is `two-zero`. $Y = T = \|S^3\|_4$ (Proposition 4): Theorem 31 at T, with the full Lemma 13.

Theorem 33 ($\pi_4(S^3) = \mathbb{Z}/2$)**.** *Under Axiom 1, $\pi_4(S^3) \cong \mathbb{Z}/2$ as groups.*

Proof. As `SphereUniversal.Two`, one level up, on classes only. $u \in [T, BR]_*$ with $[\eta]; u = s$ (η^* at BR, $\mathsf{isTrunc}\,6$); $t \in [BR, T]_*$ with $s; t = [\eta]$ (Theorem 32 at T). Then $s; (t; u) = [\eta]; u = s = s; \mathrm{id}$, so $t; u = \mathrm{id}_{BR}$ (Theorem 32 at BR, injective); and $[\eta]; (u; t) = s; t = [\eta] = [\eta]; \mathrm{id}$, so $u; t = \mathrm{id}_T$ (η^* at T). π_4 of a class a is $\pi_3 C(\Omega C\, a)$, π_3 of the looped class. Obligations: ΩC, looping on classes (`STr.Lift` of `Maps.loop`; the target `HC` is a set); its functoriality on classes (`loop-comp`, `loop-id`); $\pi_3 C$ of a class is a homomorphism (`Pi3C`, already); and a level-4 interface `Trunc4` and `Two4`, replacing `Interfaces.Trunc3` and the hypothesis `isTrunc five P` of `Two`. So $\pi_4 u$ and $\pi_4 t$ are inverse isomorphisms. Then

$$\pi_4 S^3 \cong \pi_4 T \cong \pi_4 BR = \pi_3(\Omega BR) \cong \pi_3 R \cong \mathbb{Z}/2$$

(Proposition 4; the above; by definition; $\pi_3\phi$ for the pointed equivalence ϕ, `gi-of-equiv`; `Triple[Zab,Z2ab,kbar]`. □

11. Obligations, and what to check

New mathematics (to be refereed): Lemma 13 (necessity, weak form), Proposition 12 with Lemma 10 (second-variable naturality of $\mathcal{E}$, the one genuinely new computation; its pointing step is flagged in the proof), Proposition 22 (the restricted lifting argument), Proposition 17.

Same text, indices raised: Lemmas 2, 3, Proposition 4 (`SphereAxiom`, `SphereTruncation`); Theorem 8 (`MapFib` §2); Lemmas 20, 21, Corollaries 23, 25, Theorem 29 (`MapFib` §§3–5); Proposition 18 (`FibreH3` with a fibre coefficient group); Theorem 31 (`Presentation`); Theorem 33 (`SphereUniversal.Two`).

Settled, and not claimed. The numerical converse of Proposition 21 of `doc/cross-term.pdf` (if $\mathrm{conj}(\kappa; K(h,4))$ deloops then $2h = 0$) is proved here (Corollary 15). Still open, and not used: the route through a normalised cell of `lemma9`, and the general cross-term formula. Lemma 13 uses only pointwise information about $\Omega\hat{G}$. No group structure on $[K_2, K_4^B]_*$ is used ($4h = 2h$ is an equation in $\mathrm{Hom}(\mathbb{Z}, B)$). No representative of a class is chosen.

Points for the referee. (1) Lemma 10: step (2) uses Lemma 9 to remove the conjugation by the pointing of $G^\sharp$, and step (3) loops an *unpointed* homotopy through `LoopHomotopy` and Eckmann–Hilton, as `ENat.e-nat`. (2) Lemma 13(i): the identification $\mathrm{conj}(K(2,2); H) = \mathsf{sq}; \mathrm{conj}H$ needs conj of a composite with *different* groups on the two sides (e_2 of $\mathbb{Z}$, e_4 of B) and the locked `Lev` versions of e_m, $K(2,m)$. (3) Proposition 22: that $\mathsf{Unloop}(\Omega M)$ is a proposition requires `Levels[2,1]` at target K_5^B ($\mathsf{isTrunc}\,7$), as in `DeloopR`. (4) Theorem 31(4): the level of `cor5-equiv`. (5) Whether $\pi_4 := \pi_3 \circ \Omega$ agrees with the group structure intended by the final statement (it is $\pi_1(\Omega^3)$ by definition of `pi3`).

12. Proposed correspondence

- `Sphere3Axiom`: `ev3`, `Sphere3[S3,base,surf,ump3]`: `evPt`, `pointed-up`, `evPt-nat` (Lemma 2).
- `Sphere3Trunc`: `s3-conn` (Lemma 3), `Trunc4` record and `trunc4` (Proposition 4); `pi4`, `pi4-map`, `loopC`, `pi4C` in `HomotopyGroups` style.
- `Necessity`: `ccr-ev` (Lemma 9), `E2`, `A1`, `A2`, `hh`, `e-pre` (Lemma 10), `iota-dbl` (Lemma 11), `quad`, `kappa-dbl` (Proposition 12), `N.pointwise`, `N.cls`, `N.two-tors` (Lemma 13), `conv` (Corollary 15), `factor` (Lemma 16), `Psi`, `psi-phi`, `psi-inj` (Proposition 17).
- `FibreH3`: `FibAtC[C,k,inj,B]` (the existing proof with a fibre coefficient group); `fib-h3-R` (Proposition 18).
- `MapBR`: `Triple3[A,B,K]` (fibre Fib$K$, its pointing, `conn4` via `FibConn`, `trunc6`, connectedness of $\Omega\mathrm{Fib}K$), `Conj[eA,eB]` (Conventions), `Lifting3` (Theorem 8); `SourceBR`: `cp`, `PStar`, `fseq`, `ExBR`, `ex-BR` (Proposition 22), `un-BR` (Proposition 19), `qBR`; `SourceS3`: `s3-paths`, `qS`, `sigma3`, `Sig3` (Definition 6, Proposition 7); `Final`: `s`, `s-eq`, `right`, `rep-fib`, `rep-pteq`.
- `Presentation4`: `Rec4`, `Pres4[Y,y0,cY,tY]`, `pres4` (Theorem 31).
- `Pi4Sphere`: `up` (Theorem 32), `Two4`, `pi4-S3 : GIso[pi4 S3 base, ab-grp Z2ab]` (Theorem 33).

Each declaration gets an entry in a new spec file (`br.opens`), every line "nothing else opened".